Provisional Patent Application 23 — Liana Banyan Corporation

Cooperative-Class Campaign Mechanics, CSIA Distributed Inference, and Stamp-Certified IP Attribution — 7 Claim Groups — Provisional 23

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Provisional Patent Application 23

Liana Banyan Corporation

Applicant: Liana Banyan Corporation **Entity Type:** Wyoming C-Corporation (Small Entity) **EIN:** 41-2797446 **Inventor:** Jonathan Jones (sole inventor) **Filing Type:** Provisional Patent Application (USPTO Patent Center) **Estimated Filing Fee:** $65 (Small Entity, provisional) **Priority Date Goal:** [to be completed by Applicant at USPTO Patent Center] **Conversion Deadline:** 12 months from filing date **Filing path:** USPTO Patent Center direct — Founder discretion.

Title of Invention: *System and Method for Cooperative-Class Campaign Mechanics, Distributed Inference Architecture with Pheromone Routing, and Stamp-Certified IP Attribution in Cooperative AI Substrate Platforms*

Filed under: #2260 Cooperative Defensive Patent Pledge (Liana Banyan Corporation, BP098).

PUBLICATION GATE HARD: This specification is **internal-only** until Founder USPTO submission. Knight scope: build and stage. Founder scope: review and authorize USPTO submission. Fire Control per #2330; BP098.

Cross-References to Related Applications

This application claims priority benefit from related provisional applications, all filed by the same applicant (Liana Banyan Corporation) with the same sole inventor (Jonathan Jones):

- **Provisional Application Series 1-22**, filed 2024-2026, covering the Cooperative-Architecture AI Substrate primitives, Pheromone Substrate, Cathedral Effect, Wrasse Pre-Injection, Soccerball-DAG, Canonical Plow Pipeline, FireGuard Staggered Swarm, CSIA architecture foundational claims, and related cooperative-class innovations.

The architecture disclosed in this application composes with, extends, and provides reduction-to-practice evidence for innovations in those prior applications. This filing constitutes the integration of 7 claim groups ratified across BP085-BP098 covering CSIA Empress event-driven naming and project-funding mechanics, Bonfire Lit Scribe active peer-mesh connection, Mountain 1 priming-propagation, Cue Deck Card unified 7-facet seed-loop system, Bounty Posters cooperative-class campaign mechanics, CSIA architecture 8-angle distributed inference, and Stamp-Certified IP Ledger with Ring Bearer Merkle replication.

Most relevant prior filing:

- **Provisional Application No. 64/095,518**, filed June 21, 2026 (Provisional 22) — *System and Method for a Cooperative Local-First AI Substrate with Multi-Specialist Plow and Adversarial Verification* — 37 Claim Groups including Hexadecimal Machine Code wire format, Brain-Swap hot-swappable cognitive core, and substrate architectural primitives composing with the PROV_23 claims herein.

Filing Manifest — Innovation Cluster Summary

#	Innovation Cluster	Short Description
CG1	CSIA Empress Event-Driven Naming + Project-Funding Mechanic	Event-driven cohort threshold; 60-winner 1-per-country tournament; cooperative project funding with transparent-accounting post-payment reveal; IP Ledger attribution chain
CG2	Bonfire Lit Scribe — Active Peer-Mesh Connection Primitive	Continuous heartbeat monitoring; priming-context drift detection; 4-trigger auto-recovery; default-active on Tier 6 unlock; user-toggleable sovereignty
CG3	Mountain 1 Priming-Propagation Primitive	Orchestrator-generated cooperative substrate priming context propagated to all subordinate peer frames before query dispatch; measurable inter-peer agreement rate increase
CG4	Cue Deck Card Unified 7-Facet Seed-Loop System	Single canonical card variant-rendered for Twitter/LinkedIn/Facebook; SOCCERI network-join address transmission; tiered Marks attribution; Medallion cooperative rights; anti-MLM single-hop constraint
CG5	Bounty Posters — Cooperative-	Cooperative bounty bound to goal; event-driven cohort threshold gate; IP Ledger

#	Innovation Cluster	Short Description
	Class Participatory Campaign Mechanic	attribution binding; Loop 11 elimination verification as IP name-credit gate; Pioneer Bonus composition
CG6	CSIA Architecture — 8 Independent Patent Angles	Pheromone-trail routing (O(k)); Three Fates fan-out calibration; canonical ABSTAIN output; training integrity triple; distributed inference with Marks attribution; full provenance chain
CG7	Stamp-Certified IP Ledger + Ring Bearer + Frontier Mesh Merkle Replication	Ed25519-signed Merkle-chained local ledger per Ring Bearer; Merkle-diff replication to all Frontier Mesh peer contacts; auto-certification of open-weight AI models on pull; offline verification

Total: 7 Claim Groups — Pledge #2260

Cooperative Defensive Patent Pledge (#2260)

Filed under the LB Cooperative Defensive Patent Pledge (originated B098): *"We file the IP so no one can lock it away, and we pledge to never use it offensively against anyone building on the cooperative substrate."* Every nonprofit, cooperative, and academic institution receives use rights free and forever on IRS-verified EIN or international equivalent. Claims drafted maximally broad, defensible on Path B empirical-receipt grounds.

Path B Discipline (Proof Before Claim)

All claims are grounded in empirical receipts (Path B, B130 Founder-ratified): build the reduction-to-practice artifact first; file the provisional with the working build as Section 2 evidence. No claim in this filing is a glorified pitch deck.

Claim Group	Empirical Anchor	Call Sign / Commi
CG1 Empress	`empress_proposals` (15 cols incl. ip_ledger_hash, ip_ledger_entry_id, cohort_id, real_votes, ghost_votes, status); `empress_cohorts` (7 cols incl. cohort_uuid, prize_pool_share); `empress_votes_real` + `empress_votes_ghost` tables — dual-vote mechanic wired in DB	SEG-V BP097-BP098
CG2 Bonfire Lit	`peer_presence` table wired; Trial 7 receipt: M2+M3 abstain at cascade without priming propagation; M0 answers correctly — priming-context-diff is documented structural cause	SEG-U BP098
CG3 Mountain 1	Trial 7/8 receipt: M0 correct while M2+M3 abstained without priming; priming-context differential is the measurable before/after anchor confirming inter-peer divergence root cause	BP097-BP098

Claim Group	Empirical Anchor	Call Sign / Commi
CG4 Cue Deck	`cue_card_templates` 161 live rows confirmed 2026-06-27; live card: id `75df3887-3fc7-4f18-bf69-118043fd3809`, title "Discover the Secret of Mnem", initiative_slug `empress-naming-campaign`, is_active=true	BP093 canon ratify
CG5 Bounty Posters	`bounty_posters` (10 cols), `bounty_claims`, `bounty_contributions`, `ip_attribution_ledger`, `pioneer_bonus_ledger` (13 cols incl. pioneer_position, marks_multiplier, gauntlet_run_id) — all wired in DB schema	BP084 / BP047 Prov 20 CF6
CG6 CSIA	BP094 smoke: peer 88cbf6bd (gemma4:12b, 12B params) correct via cooperative routing; peer cb4ef450 (llama3.3:70b, 70B params) wrong without cooperative grounding — smaller cooperative model beats larger raw-parameter model	BP094 smoke receipt
CG7 IP Ledger	`ip_ledger` (10 cols incl. ring_bearer_id, ed25519_sig BYTEA, stamp_seq, merkle_node, replicated_at); `ip_ledger_entries` (20 cols incl. ed25519_sig BYTEA, mesh_replicated BOOLEAN, patent_application_number,	BP087 canon ratify

Claim Group	Empirical Anchor	Call Sign / Commi
	filing_status, csia_phase) — Ed25519 and Merkle as first-class DB schema fields	

Filing Window and Priority

This provisional application is filed June 2026 (BP098). The 12-month non-provisional conversion deadline for this filing is June 2027. The non-provisional conversion path activates upon Founder authorization at USPTO Patent Center.

Chain context: 22 provisional applications were filed by Liana Banyan Corporation preceding this filing (Prov 1 through Prov 22); this filing is the 23rd.

The sequential provisional chain protecting the cooperative-platform AI substrate architecture:

- Prov 13: 64/036,646 (April 12, 2026) — Cathedral Effect, Pheromone Substrate baseline
- Prov 14: 64/052,602 (April 29, 2026) — Multi-thousand-innovation formal ledger cluster
- Prov 15: 64/052,618 (April 29, 2026) — B132-B133 primitive cluster (Wrasse, Stone Tablet, Fire Control)
- Prov 16: Filed May 1, 2026 — BP002-BP006 architectural ratifications catch-all
- Prov 17-21: Filed 2026 — covering additional cooperative substrate primitives
- Prov 22: 64/095,518 (June 21, 2026) — BP083-BP086 Plow Cycle: 37 Claim Groups
- **Prov 23: THIS FILING (June 2026)** — BP085-BP098 Campaign Mechanics and CSIA Architecture Cluster: 7 Claim Groups

Background of the Invention

Field of the Invention

The present invention relates to artificial intelligence systems and cooperative-class participatory platform mechanics, and more particularly to: (i) event-driven cooperative naming and project-funding mechanisms that open on member participation thresholds rather than calendar schedules, structuring competitive fields with geographic diversity constraints and binding winning outcomes to tamper-evident IP attribution records; (ii) active peer-mesh connection scribes that monitor cooperative frame health and maintain priming-context synchronization across distributed compute frames; (iii) priming-context propagation methods that transmit orchestrator-generated cooperative substrate context to subordinate peer frames before query dispatch, measurably increasing inter-peer inference agreement; (iv) unified multi-platform recruitment card systems with cryptographic cooperative-identity transmission and anti-MLM structural constraints; (v) participatory bounty campaign mechanics that bind contribution artifacts to IP attribution registries through cooperative verification gating; (vi) distributed inference architectures using pheromone-trail routing in place of neural self-attention, with canonical refusal output and cooperative training data integrity verification; and (vii) stamp-certified intellectual property attribution ledgers with cryptographic per-entry signing and Merkle-proof replication across cooperative peer networks.

Background and Problem Statement

The cooperative-class platform architecture presented herein addresses multiple simultaneous limitations across the cooperative technology landscape:

Problem 1 — Cooperative Naming Without Event-Driven Threshold Mechanics. Existing naming campaigns for platforms and cooperatives operate on calendar-based schedules (e.g., a 30-day contest window regardless of participation level), producing sparse or unrepresentative participation from members who were not yet present at campaign time. No prior cooperative platform implements a naming campaign that opens only upon reaching a meaningful member participation threshold, or structures the competitive field geographically to prevent single-country dominance of the tournament. No prior platform binds the winning name to a tamper-evident IP Ledger registration with contributor attribution at the moment of selection. Further, no prior platform implements a cooperative project-funding flow requiring minimum floor payment from participants as a prerequisite to name submission, with a cooperative-fund fallback split and a transparent-accounting post-payment redirect that reveals to the member where their contribution flows after payment completion rather than before — a UX mechanic that completes the trust-building commitment before disclosing the accounting detail.

Problem 2 — Distributed Inference Frame Disconnection. In existing multi-frame AI substrate architectures, subordinate peer compute frames operate without continuous synchronization of priming context from the orchestrator frame. This produces priming-context drift: a state in which the orchestrator has been updated with current cooperative substrate context but subordinate peers have not, causing subordinate peers to answer from stale or absent context. Existing peer orchestration architectures do not detect heartbeat-stale conditions, priming-context drift events, route-phantom states, or MCP-respawn-needed conditions. They have no mechanism for recovery scripts auto-fired on these four specific trigger conditions. No existing architecture provides a user-toggleable connectivity primitive that is default-active upon a cooperative membership tier unlock event without propagating user-private data across the peer network.

Problem 3 — Multi-Peer Dispatch Without Priming Propagation. Existing multi-model inference orchestrators dispatch queries to peer models without first establishing shared context across all peer frames. The result is that peer models answer from base training or from independently-retrieved context, producing high inter-peer variance. No prior system structures multi-peer dispatch around an orchestrator-generated priming context derived from a cooperatively-ratified substrate and transmitted to all subordinate peers before query dispatch, such that inter-peer agreement rate measurably increases compared to unprimed baseline. Empirical evidence in this filing (Trial 7, BP097/BP098) shows that M2 and M3 abstain at cascade without priming propagation, while M0 answers correctly — confirming that priming-context differential is the structural root cause of inter-peer divergence.

Problem 4 — Recruitment Cards Without Cooperative Substrate Integration. Existing referral card systems generate static images or text snippets for social media sharing without binding the card to a cooperative platform's peer-mesh substrate. No prior system: stores a single canonical card record variant-rendered for multiple social platforms without creating separate records; transmits the sharing member's cryptographic cooperative-substrate identity (SOCCERI address, defined as SHA-512 of peer payload bytes in lowercase hex) as part of the invite payload so the recipient's signup connects them to the sharer's cooperative substrate network circle; enforces an anti-MLM structural constraint by capturing attribution at one hop only with no chain propagation table; or confers cooperative participation tier rights (Medallion class) from the card-sharing event itself independent of conversion.

Problem 5 — Bounty Mechanics Without IP Attribution and Loop Verification Gating. Existing crowdsourced bounty mechanics (bug bounty platforms, crowdsourcing platforms) do not bind contribution artifacts to patent IP attribution ledgers. No prior system gates name-credit on patent claims behind a Loop 11 elimination verification event; records contribution artifacts in a tamper-evident cooperative labor registry with cooperative verification evidence; or composes a transparent-accounting post-payment reveal as a trust-building UX primitive with the bounty payout event.

Problem 6 — AI Inference Without Cooperative Substrate Routing. Existing large language model inference architectures rely on: neural self-attention over statistically-learned token embeddings (O(n squared) scaling); dense matrix multiplication in feed-forward layers; single-model output sampling with no structural mechanism for refusing to answer when cooperative substrate knowledge is insufficient; and unverified training corpus ingestion without multi-peer adversarial consensus, deterministic sync verification, or cryptographic provenance. No prior architecture provides: pheromone-trail routing as a substitute for self-attention (O(k) sparse, cooperatively-ratified); a canonical ABSTAIN output class structurally preventing forced completion on low-confidence queries; a training data integrity triple (Star Chamber + Scrambler + Keys Engines) providing adversarial verification before substrate entry; or distributed inference across cooperative member nodes with member- attributed Marks accumulation per correct inference contribution.

Problem 7 — IP Ledgers Without Cryptographic Peer Replication. Existing intellectual property attribution registries are centralized: a single authoritative copy maintained by a platform operator, with no cryptographic proof available to individual contributors. If the platform operator's system fails, attribution records are at risk. No prior cooperative IP ledger architecture provides: Ed25519 signing of each entry by the local Ring Bearer keypair; Merkle-chained entry sequence enabling tamper detection on any historical entry; continuous Merkle-diff replication to every peer frame the Ring Bearer contacts; auto-certification of locally-installed open-weight AI models as IP Ledger entries upon pull with SHA256 hash verification; or offline verification of contribution entries using a local Ledger copy without platform connectivity.

Empirical Receipt

The campaign mechanics and distributed inference architecture claimed herein have been empirically demonstrated across the following receipts:

Campaign Mechanics: The `empress_proposals` database table (15 columns including ip_ledger_hash, ip_ledger_entry_id, cohort_id, real_votes, ghost_votes, status CHECK constraint) and `empress_cohorts` table (7 columns including cohort_uuid, prize_pool_share, status) establish that the event-driven cohort lifecycle is wired at the database layer, not merely conceptual. The `empress_votes_real` and `empress_votes_ghost` tables confirm the dual-vote mechanic is architecturally implemented. The `cue_card_templates` table (26 columns, 161 live rows as of 2026-06-27) confirms the Cue Deck Card unified system is in active production with the

Empress Deck Card (`empress-naming-campaign` slug, is_active=true) confirmed live. The `bounty_posters`, `bounty_claims`, `bounty_contributions`, `ip_attribution_ledger`, and `pioneer_bonus_ledger` tables confirm the Bounty Poster mechanic is architecturally wired at the database schema level.

Distributed Inference: Trial 7 (BP097/BP098) demonstrated that peer frames M2 and M3 abstain at cascade without priming propagation, while orchestrator frame M0 answers correctly. The priming-context differential is the documented structural cause of the inter-peer divergence. BP094 smoke receipt demonstrates that peer 88cbf6bd (gemma4:12b, 12 billion parameters) answers correctly via cooperative pheromone routing while peer cb4ef450 (llama3.3:70b, 70 billion parameters) answers incorrectly without cooperative grounding — confirming that cooperative substrate routing produces correct inference from a smaller model that raw parameter scale alone does not achieve.

IP Ledger: The `ip_ledger` table (10 columns including ring_bearer_id, ed25519_sig BYTEA, stamp_seq, merkle_node, replicated_at) and `ip_ledger_entries` table (20 columns including ed25519_sig BYTEA, mesh_replicated BOOLEAN, patent_application_number TEXT, filing_status TEXT, csia_phase TEXT) confirm the Ring Bearer architecture is wired at the database layer with Ed25519 signing and Merkle replication as first-class schema fields, not as future design aspirations.

Summary of the Invention

The present invention discloses innovations in the following areas, each constituting a novel and non-obvious contribution to the art of cooperative-architecture artificial intelligence and cooperative platform mechanics:

1. **CSIA Empress Event-Driven Naming + Project-Funding Mechanic** — an event-driven cohort threshold naming campaign opening only upon reaching a member participation threshold (empirically 10,000 members), structured as a rolling 3-week window with 60-winner 1-per-country geographic diversity tournament, minimum floor payment plus variable-amount cooperative project-funding contribution, transparent-accounting post-payment redirect, and IP Ledger registration of the winning name with contributor provenance chain and Loop 11 verification gate before patent name-credit.

2. **Bonfire Lit Scribe — Active Peer-Mesh Connection Primitive** — a continuously-running peer-mesh connection scribe that monitors peer heartbeats via `peer_presence` table, detects priming-context drift between orchestrator and subordinate frames, auto-fires recovery scripts on four distinct trigger conditions (heartbeat-stale, priming-drift, route-phantom, MCP-respawn), is default-active on cooperative membership Tier 6 unlock, is user-toggleable off for sovereignty preservation, and never propagates user-private data across the cooperative network — only cooperative-substrate state (priming context, route tables, model registry).

3. **Mountain 1 Priming-Propagation Primitive** — a method for cooperative distributed inference wherein an orchestrator peer frame generates priming context from the cooperative substrate (canonically-ratified eblets, session receipts, domain-primed context) and transmits it to all subordinate peer frames before query dispatch, such that subordinate frames answer from the received priming context rather than

generating independent context, and inter-peer answer agreement rate measurably increases compared to unprimed baseline dispatch. *(Founder direction required: standalone CG3 or sub-claim under Bonfire Lit (CG2) — flag for v02 if preferred as dependent claim.)*

4. **Cue Deck Card Unified 7-Facet Seed-Loop System** — a cooperative recruitment mechanism implementing a single canonical card record variant-rendered for Twitter, LinkedIn, and Facebook without creating separate database records per platform; served via cooperative substrate API to peer frames; bound to cooperative initiative_slug; implementing 7 distinct recruitment facets including SOCCERI network-join transmission of the sharing member's SHA-512 cooperative identity address; tiered Marks referral attribution rewards on a Pioneer- to-Ambassador ladder; Medallion cooperative participation rights conferred at card-share event; and single-hop anti-MLM structural constraint with no chain attribution table.

5. **Bounty Posters — Cooperative-Class Participatory Campaign Mechanic** — a participatory cooperative campaign mechanism comprising a structured bounty bound to a cooperative goal requiring content artifact contribution, event-driven cohort threshold gate, Marks distribution on verified completion, tamper-evident cooperative labor registry recording with permanent contributor name attribution, Loop 11 elimination verification as a hard gate for IP patent name-credit, and transparent-accounting post-payment reveal as a trust-building UX primitive composing with the bounty payout event.

6. **CSIA Architecture — 8 Independent Patent Angles** — a cooperative substrate inference architecture comprising: (1) cooperative-substrate-ratified token vocabulary in place of statistically-learned embeddings; (2) pheromone-trail routing in place of self-attention (O(k) sparse vs. O(n squared)); (3) cel-class hierarchical feed-forward projection via e-giant/e-sprite/e-spider topology; (4) Three Fates multi-model fan-out calibration;

5. canonical ABSTAIN output class structurally preventing forced-completion on low-confidence queries; (6) training data integrity triple (Star Chamber + Scrambler + Keys Engines);
6. distributed inference with member-attributed Marks accumulation; and (8) full provenance chain from output to catacomb row.

7. **Stamp-Certified IP Ledger + Ring Bearer + Frontier Mesh Merkle Replication** — a system for cooperative IP attribution and preservation wherein each installing member's machine (the Ring Bearer) maintains a local stamp-certified IP Ledger with Ed25519-signed Merkle-chained entries; Merkle-diff replication to every peer frame in the cooperative Frontier Mesh occurs continuously; locally-installed open-weight AI models are auto-certified as IP Ledger entries upon pull with SHA256 hash; the cooperative ledger is by design a blockchain-style distributed backup with no single authoritative copy; and any Ring Bearer can verify their contribution entries offline using their local Ledger without platform connectivity.

Detailed Description of the Preferred Embodiments

Claim Group 1: CSIA Empress Event-Driven Naming + Project-Funding Mechanic

Claim 1.1 (CSIA Empress — Independent System and Method Claim)

A computer-implemented method for cooperative-class participatory naming and project-funding comprising:

- a. monitoring a cooperative membership participation register to detect when a member participation count crosses a configurable threshold (empirically 10,000 members) rather than triggering on a fixed calendar date, such that the naming cohort opens only when a meaningful cooperative participation threshold has been achieved;
- b. upon threshold crossing, opening a rolling cohort window of configurable duration (empirically 3 weeks) for cooperative name submissions by active members, wherein the cohort closes upon winner selection rather than upon expiration of a fixed deadline;
- c. structuring the competitive field as a tournament with a configurable number of winners (empirically 60) subject to a one-per-country geographic diversity constraint such that no single country provides more than one tournament winner from any single cohort;
- d. requiring each participating member to make a cooperative project-funding contribution comprising a minimum floor payment (empirically $5) plus an optional variable-amount additional contribution,

with the total contribution directed to a cooperative fund at proportions [Founder-ratify: proportions TBD — proposed 1/3-1/3-1/3];

- e. delivering a transparent-accounting post-payment redirect that reveals to the member, after payment completion, the exact destination of each fraction of their contribution — constituting a trust-building UX mechanic in which the member has already committed before seeing the ledger accounting detail; and
- f. recording the winning name and its contributor attribution in a tamper-evident IP Ledger with a provenance chain linking the submitted name text, the submitting member identifier, the cohort event identifier, and the IP Ledger entry hash, such that the cooperative naming event produces a verifiable IP provenance record from submitted name to ledger entry.

Claim 1.2 (CSIA Empress — Dependent: Cohort Lifecycle State Machine)

The method of Claim 1.1, further comprising:

- a. maintaining a cohort lifecycle state machine with states comprising at minimum: `inactive`, `active`, `voting`, `closed`, and `finalized`, wherein the cohort transitions from `inactive` to `active` only upon the member participation threshold crossing event of step (a);
- b. storing each cohort instance in a persistent record comprising: a cohort_uuid, a prize_pool_share value, a status field subject to a CHECK constraint permitting only canonical state values, and a created_at timestamp; and
- c. persisting each name proposal in a proposals record comprising: the cohort_id foreign key, a contributor member

identifier, a real_votes count, a ghost_votes count, a status field, and at minimum one IP Ledger linkage field (ip_ledger_hash or ip_ledger_entry_id).

Claim 1.3 (CSIA Empress — Dependent: Geographic Diversity Constraint Enforcement)

The method of Claim 1.1, wherein the one-per-country geographic diversity constraint of step (c) is enforced at the winner selection stage by:

- a. tagging each submission with the submitting member's declared country of residence or primary cooperative jurisdiction at the time of submission;
- b. ranking all submissions by a cooperative vote score computed from the real_votes count of the proposals registry;
- c. iterating through the ranked list and selecting the highest-scoring submission from each country not yet represented among selected winners, continuing until the configurable winner count is reached; and
- d. storing the selected winner's country code as part of the winning submission record to enable subsequent audit of geographic diversity compliance across cohorts.

Claim 1.4 (CSIA Empress — Dependent: Transparent-Accounting Reveal Protocol)

The method of Claim 1.1, wherein the transparent-accounting post-payment redirect of step (e) comprises:

- a. completing the payment transaction and confirming funds receipt before displaying any breakdown of fund destination — the accounting detail is withheld from the member until after payment is complete;
- b. generating a per-payment accounting record comprising: the total amount

received, the cooperative fund allocation at each proportion slot [Founder-ratify: proportions TBD], the designated recipient of each slot, and a timestamp;

- c. displaying this accounting record to the paying member in a post-payment confirmation screen that the member cannot access until payment is confirmed; and
- d. storing the accounting record in an auditable ledger accessible to the paying member on demand, constituting a transparency guarantee without creating a pre-commitment-disclosure barrier that could reduce participation.

Claim 1.5 (CSIA Empress — Dependent: Dual-Vote Demand-Signal Mechanic)

The method of Claim 1.1, further comprising:

- a. maintaining two parallel vote registries for each name proposal: a `real_votes` count incremented only by members whose membership status is verified active, and a `ghost_votes` count incremented by non-member accounts that have interacted with the cooperative surface without completing membership;
- b. using the `ghost_votes` count as a demand-signal metric indicating non-member interest in specific name candidates without granting voting authority to unverified accounts; and
- c. using the `real_votes` count as the authoritative ranking input for winner selection, such that ghost participation provides a demand signal for cooperative growth analytics without distorting the cooperative-member-only winner determination.

Claim 1.6 (CSIA Empress — Dependent: Loop 11 IP Attribution Gate)

The method of Claim 1.1, further comprising:

- a. subjecting each winning name to a Loop 11 elimination verification process before the name-credit is registered as a patent-class IP attribution in the cooperative IP Ledger;
- b. Loop 11 elimination verification comprising: confirming the name passes adversarial challenge from a cooperative verification agent; confirming no prior art IP conflict exists on the proposed name; and recording the verification result in the contribution’s IP Ledger entry;
- c. such that only names surviving Loop 11 produce permanent IP attribution records in the cooperative ledger, and names that fail Loop 11 revert to `eliminated` status in the proposals registry with the eliminating evidence recorded for audit.

Claim Group 2: Bonfire Lit Scribe — Active Peer-Mesh Connection Primitive

Claim 2.1 (Bonfire Lit Scribe — Independent System Claim)

A cooperative computing substrate system comprising:

- a. a plurality of peer compute frames each running a locally-installed open-weight language model and connected to a cooperative peer-mesh network;
- b. an active scribe process running continuously on an orchestrator frame that monitors peer heartbeats by polling a `peer_presence` persistence store at a configurable interval, wherein a heartbeat entry older than a configurable staleness threshold indicates a heartbeat-stale condition;

- c. a priming-context fingerprint stored on the orchestrator frame representing the current state of the cooperative substrate's session context, wherein each subordinate peer frame also stores a local priming-context fingerprint;
- d. a priming-context drift detector that compares the orchestrator's priming-context fingerprint to each peer frame's fingerprint and detects drift when the fingerprints diverge beyond a configurable threshold;
- e. a recovery script dispatcher that auto-fires one of four recovery scripts based on detected trigger condition: (i) `heartbeat-stale-recovery` on heartbeat-stale condition;
  - ii. `priming-drift-recovery` on priming-context drift; (iii) `route-phantom-recovery` on detection of a route table entry pointing to a peer frame that is no longer reachable;
  - iii. `mcp-respawn-recovery` on detection that a peer frame's MCP server process has exited and requires restart;
- f. a capability-tier gate such that the active scribe process enters default-active state only upon the cooperative membership Tier 6 unlock event associated with the $5 membership payment completing the cooperative onboarding Gauntlet; and
- g. a user sovereignty toggle that allows any member to disable the active scribe process on their frame without affecting other peer frames' scribe processes, without requiring authorization from any cooperative platform operator, and without affecting the member's cooperative participation status or Marks accrual.

Claim 2.2 (Bonfire Lit Scribe — Dependent: Heartbeat Monitor Schema)

The system of Claim 2.1, wherein the heartbeat-stale detection of step (b) further comprises:

- a. a `peer_presence` table schema comprising at minimum: a peer_frame_id primary key, a last_heartbeat_at timestamp, a current_model_name field, and a priming_context_fingerprint field;
- b. a heartbeat emitter on each peer frame that writes to the `peer_presence` table at a configurable heartbeat interval, updating the last_heartbeat_at and current_model_name fields; and
- c. a staleness check on the orchestrator that queries `peer_presence` and flags any peer whose last_heartbeat_at is more than N times the heartbeat_interval milliseconds in the past, where N is a configurable integer multiplier (empirically N=3).

Claim 2.3 (Bonfire Lit Scribe — Dependent: Priming-Context Propagation)

The system of Claim 2.1, further comprising:

- a. a priming-context propagation protocol wherein, upon detecting priming-context drift in a subordinate peer frame, the orchestrator: (i) serializes the current cooperative substrate priming context into a propagation payload; (ii) transmits the propagation payload to the drifted peer frame via a cooperative MCP transport; (iii) waits for acknowledgment from the peer frame confirming receipt and application of the priming context; and (iv) updates the peer frame's `priming_context_fingerprint` field in the `peer_presence` table;
- b. wherein the propagation payload contains only cooperative-substrate state (priming context, route tables, model registry entries) and explicitly excludes any user-private data, user session content, or

member-specific personal information — constituting a structural privacy boundary on the Bonfire Lit Scribe propagation channel.

Claim 2.4 (Bonfire Lit Scribe — Dependent: Route-Phantom Recovery)

The system of Claim 2.1, wherein the route-phantom-recovery script of step (e)(iii) comprises:

- a. scanning the cooperative frame's route table for entries pointing to peer frames whose `peer_presence` heartbeat has expired beyond the staleness threshold;
- b. marking phantom route entries with a `phantom` status rather than deleting them, preserving the historical routing record for audit;
- c. attempting a direct connectivity probe to each phantom route target to determine if the frame is reachable via an alternative transport path; and
- d. updating the route table entry status to `confirmed-phantom` if the probe fails, or restoring it to `active` if the probe succeeds, logging the recovery event to the cooperative recovery log in either case.

Claim 2.5 (Bonfire Lit Scribe — Dependent: MCP-Respawn Recovery)

The system of Claim 2.1, wherein the mcp-respawn-recovery script of step (e)(iv) comprises:

- a. detecting that a peer frame's MCP server process has exited by monitoring the process registry or MCP server health endpoint;
- b. issuing a respawn command to restart the MCP server process on the affected peer frame;
- c. waiting for the respawned MCP server to pass a health check before resuming

cooperative dispatch to that peer frame; and

- d. re-propagating the current priming context to the respawned peer frame upon successful health check, ensuring the newly-respawned peer is primed before receiving its first cooperative dispatch after restart.

Claim 2.6 (Bonfire Lit Scribe — Dependent: Privacy Boundary Structural Enforcement)

The system of Claim 2.1, further comprising:

- a. a data classification filter applied to all priming-context propagation payloads before transmission that inspects each field of the cooperative priming context;
- b. wherein any field classified as user-private — including but not limited to: user session transcripts, member personal information, member payment records, member health records, or member private notes — is stripped from the propagation payload before transmission to any peer frame;
- c. such that the cooperative peer-mesh connection primitive propagates only cooperative- substrate state and provides a structural guarantee that user-private data never crosses peer-frame boundaries via the Bonfire Lit Scribe mechanism, even if the member has not engaged the user sovereignty toggle of Claim 2.1(g).

Claim Group 3: Mountain 1 Priming-Propagation Primitive

[Founder direction surfaced per BP098 M42 brief: Mountain 1 has been drafted as standalone Claim Group 3 (CG3). If Founder prefers Mountain 1 restructured as dependent claim(s) under Bonfire

Lit Scribe (CG2), flag for v02 revision before filing. Standalone CG3 maximizes independent claim breadth.]

Claim 3.1 (Mountain 1 Priming-Propagation — Independent Method Claim)

A method for cooperative distributed inference comprising:

- a. an orchestrator peer frame generating a priming context by querying a cooperative substrate comprising: canonically-ratified eblet knowledge units, session receipts from prior cooperative interactions, domain-primed context strings matched to the pending query's domain classification, and cooperative authority metadata;
- b. transmitting the generated priming context to each subordinate peer frame in a cooperative constellation before dispatching any query to those peer frames;
- c. each subordinate peer frame, upon receiving the priming context, incorporating the transmitted priming context as the leading context of its inference input before executing the query;
- d. such that inter-peer answer agreement rate across the constellation is measurably higher than the baseline agreement rate when the same query is dispatched to the same peer frames without prior priming-context transmission; and
- e. recording the before/after agreement rate differential as an empirical receipt in the cooperative substrate session log, constituting a quality metric for cooperative priming effectiveness.

Claim 3.2 (Mountain 1 — Dependent: Propagation Protocol Schema)

The method of Claim 3.1, wherein the priming-context transmission of step (b) comprises:

- a. serializing the priming context as a structured payload comprising: a context_version identifier, a domain_slug field, the priming_text string, an orchestrator_frame_id, and a transmission_timestamp;
- b. transmitting the serialized payload to each subordinate peer frame via a cooperative MCP transport before any query dispatch to those peers;
- c. awaiting an acknowledgment message from each peer frame confirming receipt and application of the priming context before initiating query dispatch; and
- d. logging the transmission event with transmitted payload hash, acknowledgment timestamp per peer, and resulting per-peer priming-context fingerprint in the `peer_presence` table.

Claim 3.3 (Mountain 1 — Dependent: Agreement Rate Measurement Protocol)

The method of Claim 3.1, wherein the agreement rate measurement of step (d) comprises:

- a. dispatching an identical query to at least two subordinate peer frames in two conditions:
  b. without prior priming-context transmission as the baseline condition; and (ii) with prior priming-context transmission as the primed condition;
- b. comparing the answers from all peer frames in each condition using a canonical answer comparison function that classifies each pair of answers as: `AGREE`, `PARTIAL_AGREE`, or `DISAGREE`;
- c. computing per-condition agreement rates as the fraction of peer-frame pairs in the `AGREE` or `PARTIAL_AGREE` classification; and

- d. recording the agreement rate differential (primed rate minus baseline rate) as a cooperative substrate quality metric in the session log.

Claim 3.4 (Mountain 1 — Dependent: Cooperative Substrate Context Derivation)

The method of Claim 3.1, wherein the priming context generation of step (a) comprises:

- a. retrieving from the cooperative eblet store the N highest-salience eblets matching the query's domain classification, ranked by pheromone trail strength;
- b. retrieving session receipts from the prior K cooperative interactions in the current session that share domain overlap with the pending query;
- c. composing the priming context string by concatenating: the top-N eblet summaries, the K session receipt summaries, and a cooperative authority statement identifying the substrate version and ratification date; and
- d. such that the priming context is derived exclusively from cooperatively-ratified knowledge entries and does not incorporate unverified external sources.

Claim 3.5 (Mountain 1 — Dependent: Distinction from Prompt Injection)

The method of Claim 3.1, wherein the priming-context transmission of step (b) is structurally distinguished from prompt injection in that:

- a. the priming context originates from a cooperatively-ratified substrate, not from user-provided input or user-visible prompt strings;
- b. the priming context is applied at the cooperative coordination layer before user

query dispatch, not appended to a user-visible input context;

- c. the transmission occurs via a cooperative MCP transport channel, not via the inference model's visible conversational input; and
- d. the priming context is subject to cooperative substrate quality gates (Loop 10/11/12 verification) before it is eligible for inclusion in a priming payload, such that only cooperatively-verified knowledge reaches subordinate peer frames via this channel.

Claim 3.6 (Mountain 1 — Dependent: Multi-Session Priming Continuity)

The method of Claim 3.1, further comprising:

- a. persisting the most recent successful priming-context payload per peer frame in a local priming context store on the orchestrator frame;
- b. upon session resumption after an interruption, comparing each peer frame's current priming-context fingerprint to the persisted payload's fingerprint;
- c. re-transmitting the priming-context payload to any peer frame whose current fingerprint does not match the persisted payload fingerprint; and
- d. such that cooperative priming is automatically restored after session interruption without requiring the member to manually re-establish priming context for each peer frame.

Claim Group 4: Cue Deck Card Unified 7-Facet Seed-Loop System

Claim 4.1 (Cue Deck Card — Independent System and Method Claim)

A cooperative recruitment system comprising:

- a. a canonical card record stored in a cooperative substrate database comprising at minimum: a unique card identifier, an `initiative_slug` field binding the card to a specific cooperative initiative, a `template_type` classification, platform-specific rendered text fields for at least three distinct social platforms (`twitter_text`, `linkedin_text`, `facebook_text`), a `social_unlock_type` field, `clicks_per_frame_unlock` and `total_clicks_for_unlock` integer fields, an `is_active` boolean flag, and a `linked_deck_card_id` foreign key for card composition;
- b. a variant-rendering engine that produces platform-specific output from the single canonical card record without creating separate database records per platform, such that all platform variants are derived from and remain synchronized with the single canonical record;
- c. a cooperative substrate API that serves the canonical card and its platform variants to any peer frame in the cooperative network upon request, such that the same card is uniformly available across all cooperative surface entry points;
- d. a referral attribution chain recording the `introducer_user_id` of the sharing member on each card-share-to-membership conversion event;
- e. a tiered Marks reward structure awarding decreasing amounts per additional referral from the same sharing member, following at minimum the ladder: Pioneer 10,000,000 Marks, Vanguard 5,000,000 Marks, Pathfinder 3,000,000 Marks, Trailblazer 2,000,000 Marks, Guide 1,500,000 Marks, Ambassador 1,000,000 Marks per converted referral;

- f. a Facet (g) SOCCERI network-join mode that transmits the sharing member's SOCCERI address — defined as a SHA-512 hash of the peer payload bytes in lowercase hexadecimal — as part of the invite payload so that the recipient's cooperative membership registration connects them to the sharer's cooperative substrate network circle; and
- g. an anti-MLM structural constraint enforced by: capturing referral attribution at one hop only (single `referrer_id` column per conversion record, no chain attribution table), applying a diminishing Marks multiplier as the referrer's cumulative referral count grows per the tier ladder of step (e), and providing no mechanism for upstream propagation of referral credit to the referrer's introducer.

Claim 4.2 (Cue Deck Card — Dependent: 7 Recruitment Facets)

The system of Claim 4.1, wherein the canonical card supports at minimum seven distinct recruitment facets selectable per card instance via the `template_type` and `social_unlock_type` fields:

- a. Facet (a): value-proposition social share — presents the cooperative initiative's core value proposition with a joining call to action;
- b. Facet (b): milestone announcement — announces a cooperative participation milestone with an invitation to join before the next threshold;
- c. Facet (c): role-specific recruitment — presents a specific cooperative role or skill area and invites members with that capacity to join;
- d. Facet (d): initiative-specific — bound to a specific Sweet Sixteen initiative_slug and

presents that initiative's unique member value;

- e. Facet (e): Marks earning announcement — presents the Marks reward structure to prospective members as a recruitment incentive;
- f. Facet (f): QR-code-mediated physical invite — presents a QR code for physical-space recruitment with referral attribution preserved through the QR payload; and
- g. Facet (g): SOCCERI network-join — transmits the sharing member's SHA-512 SOCCERI address as part of the invite payload as described in Claim 4.1(f).

Claim 4.3 (Cue Deck Card — Dependent: Medallion Cooperative Participation Rights)

The system of Claim 4.1, further comprising:

- a. a Medallion system that confers cooperative participation rights from the card-sharing event itself, independent of whether the recipient converts to membership;
- b. at least four Medallion classes: CHALK_ONE_PRIME (multiplier 3x), CHALK_ONE_BUILDER (multiplier 1.5x), CHALK_ONE_SPONSORED (multiplier 1x), and SEEDLING_SPONSOR (multiplier 2x);
- c. a Medallion assignment rule binding each card template_type to a specific Medallion class, such that a sharing member earns the associated Medallion class upon a card-share event completing with a registered engagement; and
- d. the Medallion record stored in the cooperative substrate with the sharing member's identifier, the Medallion class, the multiplier value, and the card identifier that triggered the award, constituting a cooperative participation rights grant

traceable to the originating card share event.

Claim 4.4 (Cue Deck Card — Dependent: Frame Unlock Accumulator)

The system of Claim 4.1, further comprising:

- a. a frame-unlock accumulator that counts engagement events (clicks, card views, referral conversions) per card per peer frame;
- b. wherein the card's `clicks_per_frame_unlock` field specifies the per-frame engagement threshold and `total_clicks_for_unlock` specifies the aggregate engagement threshold;
- c. upon either per-frame or aggregate threshold crossing, a cooperative capability unlock event fires on the sharing member's frame, granting access to a designated cooperative capability or expanding the member's Marks earning rate for the associated initiative; and
- d. the unlock event recorded in the cooperative substrate as a capability grant with the triggering card identifier, the threshold type crossed, and the timestamp.

Claim 4.5 (Cue Deck Card — Dependent: Cooperative Substrate API Serving Protocol)

The system of Claim 4.1, wherein the cooperative substrate API of step (c) comprises:

- a. a card-serving endpoint accepting an initiative_slug parameter that delivers the canonical card record plus all platform variants in a single response payload;
- b. a peer-frame authentication check confirming the requesting peer frame is a registered cooperative member before serving active cards;

- c. a card activation state check confirming the card’s `is_active` field is true before serving, and logging a null response with a reason code for inactive card requests; and
- d. a linked-deck composition resolver that follows the `linked_deck_card_id` foreign key chain to return any card a given card is linked to in a single API response, enabling multi-card deck compositions to be resolved in a single round trip.

Claim 4.6 (Cue Deck Card — Dependent: SOCCERI Address Transmission Protocol)

The system of Claim 4.1, wherein the SOCCERI network-join transmission of step (f) comprises:

- a. the sharing member’s SOCCERI address computed as SHA-512(peer_payload_bytes) in lowercase hexadecimal, where peer_payload_bytes is the canonical serialization of the sharing member’s cooperative peer frame identity;
- b. the SOCCERI address embedded in the card’s invite URL as a URL parameter or QR code payload such that the address travels with the invite independent of the sharing platform;
- c. upon recipient membership registration, the recipient’s cooperative onboarding process reads the SOCCERI address from the invite payload and stores a bidirectional network-join record linking the sharer’s cooperative identity to the recipient’s cooperative identity; and
- d. the network-join record stored in the cooperative substrate enabling the orchestrator to route cooperative dispatch tasks to the sharer’s peer network circle, compounding cooperative network density with each SOCCERI-network-join-mode card converted to membership.

Claim Group 5: Bounty Posters — Cooperative-Class Participatory Campaign Mechanic

Claim 5.1 (Bounty Posters — Independent Method Claim)

A computer-implemented method for cooperative-class participatory campaign mechanics comprising:

- a. publishing a structured bounty record in a cooperative database comprising at minimum: a bounty_slug identifier, a bounty_title, a category field, a status field, a bp_origin reference, a goes_live_at timestamp, and a closed_at timestamp, wherein the bounty is bound to a specific cooperative goal or required content artifact type;
- b. accepting structured contribution artifacts from cooperative members in response to the published bounty, wherein each artifact is a distinct content contribution (such as a proposed cooperative name, a code commit with test evidence, an adversarial test case, or a patent claim draft) recorded with the contributing member's identifier and submission timestamp;
- c. routing each submitted artifact through an event-driven cohort threshold gate that accumulates contributions without closing on a fixed date, closing only when a participation threshold or a cooperative verification event is reached;
- d. distributing Marks — a cooperative-class participation currency — to each verified-complete contribution, sourced from the cooperative Marks treasury and recorded in the bounty_claims registry;
- e. recording all completed contributions in a tamper-evident cooperative labor registry comprising at minimum: bounty_claims,

bounty_contributions, and ip_attribution_ledger tables, with permanent contributor name attribution in each record; and

- f. subjecting each contribution artifact to Loop 11 elimination verification as a prerequisite for the contributor's name to be registered on any resulting patent claim derived from the bounty, wherein Loop 11 comprises adversarial challenge plus elimination evidence recording as described herein.

Claim 5.2 (Bounty Posters — Dependent: IP Ledger Attribution Binding)

The method of Claim 5.1, further comprising:

- a. for each contribution surviving Loop 11 verification, generating an IP Ledger attribution record comprising: the contributor's member identifier, the bounty_slug, the contribution artifact hash, a timestamp, and the cooperative initiative the contribution serves;
- b. signing the IP Ledger attribution record with the Ring Bearer's Ed25519 keypair at the time of recording;
- c. chaining the signed attribution record into the Ring Bearer's local Merkle-chained IP Ledger; and
- d. replicating the signed attribution record to peer frames in the Frontier Mesh via the Merkle-diff replication protocol of Claim Group 7, establishing a distributed backup of the contributor's IP attribution evidence across all peer frames the Ring Bearer contacts.

Claim 5.3 (Bounty Posters — Dependent: Transparent-Accounting Reveal Composition)

The method of Claim 5.1, further comprising:

- a. for bounty poster instances requiring a payment-class contribution (crowdfunded bounty prizes), composing the transparent-accounting post-payment reveal mechanic of Claim 1.4 with the bounty payout event;
- b. such that the bounty participant sees the breakdown of their payment destination (cooperative fund, prize pool, platform cost) only after payment is confirmed, constituting a trust-building UX primitive composing with the cooperative campaign mechanic; and
- c. the per-bounty accounting record stored in the cooperative substrate alongside the bounty_claims registry entry for the same payment event, enabling audit of both contribution attribution and payment flow for each bounty participant.

Claim 5.4 (Bounty Posters — Dependent: Marks Distribution with Pioneer Bonus)

The method of Claim 5.1, wherein the Marks distribution of step (d) comprises:

- a. a verification function confirming the contribution artifact meets the bounty's stated completion criteria before Marks are disbursed;
- b. a Marks distribution record written to the cooperative Marks treasury comprising: the contributor member identifier, the bounty_slug, the Marks amount, the verification event identifier, and a distribution timestamp;
- c. a pioneer bonus multiplier applied to early contributors per the Pioneer Bonus tiered multiplier system, wherein the first contributor receives the highest multiplier and subsequent contributors receive decreasing multipliers per a configurable tier table; and

- d. all distribution records stored in the `pioneer_bonus_ledger` table with pioneer_position, marks_multiplier, and gauntlet_run_id fields, enabling audit of contribution ordering and bonus amounts per bounty event.

Claim 5.5 (Bounty Posters — Dependent: Loop 11 Elimination Verification Gate)

The method of Claim 5.1, wherein the Loop 11 elimination verification of step (f) comprises:

- a. generating an adversarial challenge prompt from the contribution artifact that attempts to identify factual, logical, or IP-conflict errors in the contributed content;
- b. routing the challenge to a cooperative verification agent comprising at minimum one independent cooperative peer frame with verified domain competence in the bounty's subject area;
- c. recording the challenge result as one of: `PASS` (no eliminating evidence found), `CHALLENGE` (eliminating evidence found, with specific evidence cited), or `INSUFFICIENT_EVIDENCE` (insufficient basis to eliminate or pass);
- d. updating the contribution's status in the ip_attribution_ledger to `loop11_passed` on PASS, `loop11_eliminated` with evidence reference on CHALLENGE, or `loop11_pending` on INSUFFICIENT_EVIDENCE; and
- e. routing only `loop11_passed` contributions to the patent claim attribution registration step, such that Loop 11 is a hard gate — not a soft recommendation — for IP patent name-credit on any claim derived from the bounty contribution.

Claim 5.6 (Bounty Posters — Dependent: Pioneer Position Registry)

The method of Claim 5.1, further comprising:

- a. assigning a pioneer position integer to each contribution in chronological order of verification-confirmed submission, where position 1 designates the first verified contribution to the bounty;
- b. looking up the tiered Marks multiplier for the assigned pioneer position from the cooperative Pioneer Bonus tier table;
- c. computing the Marks award as: base_bounty_marks multiplied by pioneer_multiplier, where base_bounty_marks is a configurable per-bounty amount and pioneer_multiplier is the tier-table value for the contributor's pioneer position; and
- d. recording pioneer_position, marks_multiplier, and gauntlet_run_id as a row in the `pioneer_bonus_ledger` table, linking the pioneer bonus record to the gauntlet run that triggered the bounty's cooperative verification event.

Claim Group 6: CSIA Architecture — 8 Independent Patent Angles

Claim 6.1 (CSIA Architecture — Independent System Claim: Core Cooperative Inference)

A system for cooperative-substrate artificial intelligence inference comprising:

- a. a pheromone-trail routing index comprising cooperatively-ratified knowledge entries (eblets) each associated with a pheromone strength value that decays over time per a configurable half-life parameter and strengthens upon access, wherein inference routing traverses the pheromone index at O(k) complexity where k is the number of trails above a minimum strength threshold, rather than O(n squared) neural self-attention over a full dense embedding sequence;

- b. a cel-class hierarchical projection layer comprising e-giant nodes (domain authorities), e-sprite nodes (sub-domain specialists), and e-spider nodes (atomic knowledge units), wherein feed-forward projection traverses the cel hierarchy rather than applying dense matrix multiplication across a fixed neural architecture;
- c. a Three Fates fan-out calibration stage comprising three independent voter inference calls at distinct temperature or sampling parameters, wherein the final output is determined by concordance across the three voter results rather than single-model sampling;
- d. a canonical ABSTAIN output class that is produced when no pheromone trail in the routing index meets a minimum strength threshold, structurally preventing the system from generating a forced-completion answer on queries for which the cooperative substrate has insufficient verified knowledge;
- e. a training data integrity triple comprising three distinct verification engines: Star Chamber (multi-peer adversarial consensus verification), Scrambler (deterministic synchronization and deduplication), and Keys Engines (cryptographic provenance verification), all three required to pass before a knowledge entry is inducted into the cooperative substrate; and
- f. a provenance chain from each inference output to the originating catacomb row comprising the contributing member's original knowledge contribution, enabling audit of the cooperative knowledge source for every inference output produced by the system.

Claim 6.2 (CSIA — Dependent: Cooperative Vocabulary, Angle 1)

The system of Claim 6.1, wherein the inference vocabulary is derived from:

- a. a cooperative-substrate-ratified token vocabulary comprising terms and concepts that have passed cooperative verification, rather than a statistically-learned embedding space derived from an unverified training corpus;
- b. each vocabulary entry associated with an eblet identifier, a pheromone strength value, a ratification timestamp, and at least one Ring Bearer signature from the contributing member's Ed25519 keypair; and
- c. the vocabulary treated as a first-class IP Ledger asset, such that the cooperative token vocabulary is itself a cooperatively-owned intellectual property artifact distinct from any commercial language model's embedding space.

Claim 6.3 (CSIA — Dependent: Pheromone vs. Self-Attention Distinction, Angle 2)

The system of Claim 6.1, wherein the pheromone-trail routing of step (a) is structurally distinguished from neural self-attention in that:

- a. pheromone trail strengths are updated by cooperative member access events, not by gradient descent on a training corpus;
- b. pheromone routing complexity scales as O(k) where k is the number of high-salience trails above a strength threshold, rather than O(n squared) where n is the full sequence length;
- c. pheromone decay is governed by a configurable half-life parameter independently set per cooperative domain, not by learned attention weights; and

- d. trails that have not been accessed within the half-life period fall below the routing threshold and contribute no inference cost, providing automatic pruning of stale cooperative knowledge paths without retraining.

Claim 6.4 (CSIA — Dependent: Cel-Class Hierarchical Projection, Angle 3)

The system of Claim 6.1, wherein the cel-class hierarchy of step (b) comprises:

- a. e-giant nodes representing cooperative domain authorities, each e-giant owning a domain scope and a set of child e-sprite nodes;
- b. e-sprite nodes representing sub-domain specialists, each responsible for a bounded knowledge area within the parent e-giant's domain;
- c. e-spider nodes as atomic knowledge units (eblets) owned by e-sprite nodes, each e-spider containing a single cooperatively-verified knowledge claim;
- d. inference projection traversing from query domain classification to e-giant selection to e-sprite routing to e-spider retrieval, at each level selecting the highest-pheromone- strength nodes above the threshold; and
- e. the cel hierarchy maintained as a live cooperative substrate artifact updated on each Plow cycle, rather than as a static frozen neural architecture.

Claim 6.5 (CSIA — Dependent: Three Fates Fan-Out Calibration, Angle 4)

The system of Claim 6.1, wherein the Three Fates fan-out calibration of step (c) comprises:

- a. dispatching the same inference query in parallel to at least three voter agents at distinct sampling parameters (empirically: temperatures 0.0, 0.2, and 0.4);

- b. collecting all three voter outputs and classifying the agreement pattern as: CONCORDANT (all three agree), PARTIAL (majority agree), or DISCORDANT (no clear majority);
- c. returning the CONCORDANT or majority-PARTIAL answer as the calibrated output; and
- d. on DISCORDANT result, triggering a Diagnosis escalation rather than returning a random selection from the voter set, ensuring no forced-completion output on genuinely uncertain queries.

Claim 6.6 (CSIA — Dependent: Cooperative ABSTAIN Canonical Output, Angle 5)

The system of Claim 6.1, wherein the ABSTAIN canonical output class of step (d) is:

- a. a first-class output type with a distinct output code distinguishable from any natural-language answer string;
- b. triggered when the pheromone routing index finds no trail meeting the minimum strength threshold AND the Three Fates voter set produces a DISCORDANT result;
- c. accompanied by a structured ABSTAIN record comprising: the query that triggered ABSTAIN, the strongest available pheromone trail strength found, the Three Fates result code, and a timestamp;
- d. the ABSTAIN record written to the cooperative substrate as a Diagnosis entry, making the unanswered question available for cooperative Bounty Poster publication; and
- e. distinguished from a forced-completion hallucination in that ABSTAIN is structurally produced by the routing layer's

absence of a qualifying trail, not by the language model's confidence score or post-hoc output filtering.

Claim 6.7 (CSIA — Dependent: Distributed Inference with Marks Attribution, Angle 7)

The system of Claim 6.1, further comprising:

- a. distribution of inference queries across multiple cooperative member nodes, wherein each member node contributes local compute to the Three Fates fan-out of step (c);
- b. a Marks accumulation event fired for each cooperative member node whose voter output contributed to a CONCORDANT or majority-PARTIAL result in the Three Fates fan-out;
- c. the Marks award amount configurable per query domain and per cooperative session, reflecting the cooperative substrate's value-of-contribution principle; and
- d. Marks award records stored in the cooperative Marks treasury with the contributing member identifier, the query identifier, the voter role played, and the Marks amount awarded.

Claim 6.8 (CSIA — Dependent: Provenance Chain + Training Integrity Triple, Angles 6 and 8)

The system of Claim 6.1, wherein the provenance chain of step (f) and the training data integrity triple of step (e) together comprise:

- a. each knowledge entry inducting into the cooperative substrate bearing: a Star Chamber consensus record (multi-peer adversarial review result), a Scrambler deterministic sync hash confirming deduplication, and a Keys Engines cryptographic provenance signature linking the entry to the originating member contribution;

- b. the provenance chain from inference output comprising: output text to the eblet(s) that contributed to the output, to the catacomb row of the originating member contribution, to the originating member's Ring Bearer identifier;
- c. such that any inference output can be audited to its cooperative knowledge source, any cooperative source can be audited to the contributing member, and any knowledge entry can be verified as having passed all three components of the training integrity triple before substrate induction; and
- d. the entire provenance chain stored in the cooperative substrate as a first-class audit artifact queryable by any cooperative member, constituting a full accountability chain from inference output to member attribution.

Claim Group 7: Stamp-Certified IP Ledger + Ring Bearer + Frontier Mesh Merkle Replication

Claim 7.1 (Stamp-Certified IP Ledger — Independent System Claim)

A system for cooperative intellectual property attribution and preservation comprising:

- a. a local IP Ledger maintained on each cooperative member's computing device (the Ring Bearer), wherein the Ring Bearer is the member's local compute frame running the cooperative substrate;
- b. each IP Ledger entry signed by the Ring Bearer's Ed25519 keypair at the time of recording, producing a cryptographic

signature stored in the entry record as an `ed25519_sig` BYTEA field;

- c. entries in the Ring Bearer's local IP Ledger chained by a Merkle root updated with each new entry, such that any tampering with a historical entry invalidates the Merkle chain from the tampered entry forward, enabling tamper detection without a central authority;
- d. a continuous Merkle-diff replication protocol wherein each Ring Bearer computes the Merkle-diff between its local Ledger and each peer frame it contacts in the Frontier Mesh, and propagates entries present in the local Ledger but absent from the peer frame's replica, and vice versa;
- e. auto-certification of locally-installed open-weight AI models as IP Ledger entries upon model pull, comprising: model_name (canonical string), SHA256 hash of the model file, pull_timestamp, source_url, and Ring Bearer Ed25519 signature over these fields, stored as a first-class IP Ledger entry of type `ai_model_certification`; and
- f. an offline verification capability wherein any Ring Bearer can verify the integrity of their local IP Ledger entries using their locally-stored Ed25519 keypair and local Merkle chain without requiring connectivity to any cooperative platform service.

Claim 7.2 (Stamp-Certified IP Ledger — Dependent: Ed25519 Signing Protocol)

The system of Claim 7.1, wherein the Ed25519 signing of step (b) comprises:

- a. a Ring Bearer keypair generated once per cooperative member device and stored locally in a secure key store accessible only to the cooperative substrate process;

- b. the public key component of the Ring Bearer keypair registered in the cooperative peer network's Wrasse registry under a `public_key_hex` field, enabling any peer frame to verify the Ring Bearer's signatures without requiring the private key;
- c. each IP Ledger entry signed by: computing SHA256(entry_payload_bytes), then applying Ed25519 sign with the Ring Bearer's private key to produce the `ed25519_sig` BYTEA value; and
- d. the signature verification protocol: any peer receiving a replicated IP Ledger entry verifies the signature using the originating Ring Bearer's public key from the Wrasse registry before accepting the entry into its local replica.

Claim 7.3 (Stamp-Certified IP Ledger — Dependent: Merkle-Diff Replication Protocol)

The system of Claim 7.1, wherein the Merkle-diff replication of step (d) comprises:

- a. each Ring Bearer maintaining a Merkle tree of its local IP Ledger entries, with the Merkle root updated atomically on each new entry;
- b. upon peer frame contact, each Ring Bearer exchanging its current Merkle root with the contacted peer;
- c. if Merkle roots differ, the Ring Bearer requesting a Merkle-diff listing entries present in the peer's tree but absent in its own, and vice versa;
- d. transmitting the set-difference entries (with signatures) to the receiving peer, which verifies each signature before inserting the entry into its local replica; and
- e. such that IP Ledger replication is bandwidth-efficient (transmitting only missing entries), cryptographically verified

(signature check on every replicated entry), and eventually consistent across all peer frames in the Frontier Mesh.

Claim 7.4 (Stamp-Certified IP Ledger — Dependent: AI Model Auto-Certification)

The system of Claim 7.1, wherein the auto-certification of step (e) comprises:

- a. a model pull event hook that fires when any open-weight AI model is downloaded or pulled to the Ring Bearer's local storage;
- b. upon pull event, automatically generating an IP Ledger entry comprising: model_name (canonical string), SHA256 hash computed over the full model file bytes, pull_timestamp (UTC ISO-8601), source_url (the URL from which the model was pulled), and the Ring Bearer's Ed25519 signature over these four fields;
- c. recording the generated entry in the Ring Bearer's local IP Ledger with entry_type `ai_model_certification`; and
- d. propagating the AI model certification entry to peer frames via the Merkle-diff replication protocol of Claim 7.3, establishing distributed attestation that a specific Ring Bearer installed a specific model binary at a specific time from a specific source.

Claim 7.5 (Stamp-Certified IP Ledger — Dependent: Patent Attribution Fields)

The system of Claim 7.1, wherein the IP Ledger entry schema comprises at minimum the following patent-attribution-specific fields:

- a. `inventor_name` TEXT — the cooperative member's name as it should appear on any resulting patent attribution;

- b. `royalty_share_pct` NUMERIC — the cooperative-ratified royalty share percentage for this contribution;
- c. `cooperative_share_pct` NUMERIC — the cooperative's share percentage for this contribution;
- d. `patent_application_number` TEXT — populated upon USPTO filing, linking the IP Ledger entry to the patent application in which this contribution appears;
- e. `filing_status` TEXT — lifecycle state of the contribution's patent claim (for example: `pending_filing`, `filed_provisional`, `filed_nonprovisional`, `granted`); and
- f. `csia_phase` TEXT — the CSIA architecture phase in which this contribution was ratified, providing a temporal provenance anchor for AI-assisted contributions in the cooperative substrate.

Claim 7.6 (Stamp-Certified IP Ledger — Dependent: Distributed Backup Architecture)

The system of Claim 7.1, further comprising:

- a. by-design absence of a single authoritative IP Ledger copy: no cooperative platform server, operator, or single peer frame holds a copy designated as authoritative above any other Ring Bearer's copy;
- b. each Ring Bearer's local Ledger treated as equally authoritative, with conflicts resolved by Ed25519 signature verification — signed entries with valid Ring Bearer keys from the Wrasse registry take precedence over unsigned or invalidly-signed entries;
- c. a cooperative platform loss resilience guarantee: if the cooperative platform service becomes unavailable, all Ring Bearers retain access to their local IP

Ledger copies and can verify entry integrity offline using locally-stored keypairs; and

- d. a multi-Ring-Bearer redundancy guarantee: for any IP Ledger entry that has been replicated to N peer frames via the Merkle-diff protocol, the entry can survive the loss of up to N-1 Ring Bearer devices without being lost, providing cooperative-class distributed backup of IP attribution records.

Brief Description of Drawings

The following figures, described in text, accompany this application:

Figure 1 (described in text): CSIA Empress Event-Driven Cohort Lifecycle — state machine showing threshold crossing event triggering `inactive` to `active` transition, rolling window, voting phase, winner selection with geographic constraint, and IP Ledger registration.

Figure 2 (described in text): CSIA Empress Transparent-Accounting Post-Payment Reveal — UX flow showing payment confirmation, fund destination reveal screen, and accounting breakdown.

Figure 3 (described in text): Bonfire Lit Scribe — Active Peer-Mesh Connection Architecture — system diagram showing orchestrator frame, peer_presence heartbeat monitors, priming-context drift detector, and four recovery script dispatch paths.

Figure 4 (described in text): Bonfire Lit Scribe — 4-Trigger Recovery Script Dispatch Tree — decision tree showing heartbeat-stale, priming-drift, route-phantom, and mcp-respawn recovery script branches.

Figure 5 (described in text): Mountain 1 Priming-Propagation Protocol — sequence diagram showing orchestrator priming context generation from cooperative substrate, transmission to N peer frames, acknowledgment receipt, query dispatch, and before/after agreement rate measurement.

Figure 6 (described in text): Cue Deck Card Unified System — 7-Facet Architecture — canonical card record schema with platform variant rendering engine, initiative_slug binding, SOCCERI address transmission path, Medallion award flow, and frame-unlock accumulator.

Figure 7 (described in text): Cue Deck Card Anti-MLM Structural Constraint — single-hop attribution diagram showing referrer_id single column (no chain table), diminishing Marks tier ladder, and absence of upstream propagation path.

Figure 8 (described in text): Bounty Poster Lifecycle — state diagram showing published bounty, contribution submission, event-driven cohort gate, Loop 11 verification, Marks distribution, Pioneer Bonus assignment, and IP Ledger attribution binding.

Figure 9 (described in text): CSIA Architecture — 8-Angle Overview — architectural stack showing pheromone-trail routing (angle 2), cel-class hierarchy (angle 3), Three Fates fan-out (angle 4), ABSTAIN canonical output path (angle 5), training integrity triple (angle 6), and provenance chain (angles 7-8).

Figure 10 (described in text): CSIA Pheromone Trail Routing vs. Self-Attention — complexity comparison showing O(k) pheromone routing vs. O(n squared) self-attention, with pheromone decay curve and access-strengthening mechanic.

Figure 11 (described in text): CSIA ABSTAIN Path — decision tree showing pheromone threshold check, Three Fates DISCORDANT path, ABSTAIN output generation, and Diagnosis entry creation for cooperative Bounty Poster publication.

Figure 12 (described in text): Stamp-Certified IP Ledger — Ring Bearer Architecture — system diagram showing local ledger, Ed25519 keypair and Wrasse public key registry, Merkle chain, Frontier Mesh peer contacts, and Merkle-diff replication protocol.

Figure 13 (described in text): IP Ledger AI Model Auto-Certification Flow — event sequence showing model pull hook, SHA256 computation over model bytes, Ed25519 signing, ledger entry creation (type `ai_model_certification`), and Merkle-diff replication to Frontier Mesh peers.

Figure 14 (described in text): Cooperative IP Ledger Distributed Backup Guarantee — N-peer replication diagram showing Ring Bearer loss scenarios, N-1 redundancy guarantee, and offline verification path without platform connectivity.

Filing Gate Status

PUBLICATION GATE: HARD CLOSED

Per Fire Control Directive (#2330, Prov 23): this specification is internal-only pending Founder review and explicit USPTO Patent Center submission authorization.

Knight scope: build and stage this specification. COMPLETE. Founder scope: review this document; upload to USPTO Patent Center; authorize submission.

Estimated filing fee: **$65 (Small Entity, provisional)** USPTO Patent Center: https://patentcenter.uspto.gov

Mechanical gate: `USPTO_SUBMISSION_AUTHORIZED=false` (this specification)

Abstract

A cooperative-architecture AI platform for event-driven participatory campaign mechanics and distributed local-first inference. Empress Event-Driven Naming opens naming cohorts on member participation threshold (10,000 members), structures 60-winner 1-per-country tournaments, requires cooperative project-funding contribution with transparent-accounting post-payment reveal, and registers winning names in a stamp-certified IP Ledger. Bonfire Lit Scribe monitors peer heartbeats, detects priming-context drift, and auto-fires four-trigger recovery scripts (heartbeat-stale, priming-drift, route-phantom, MCP-respawn), default-active on Tier 6 membership unlock. Mountain 1 propagates orchestrator-generated cooperative substrate priming context to subordinate peers before dispatch, measurably increasing inter-peer agreement rate. Cue Deck Card unified 7-facet system variant-renders single canonical card for Twitter/LinkedIn/Facebook with SOCCERI network-join and anti-MLM single-hop attribution. Bounty Posters bind contributions to IP attribution ledger via Loop 11 gating. CSIA Architecture provides pheromone-trail routing, canonical ABSTAIN output, and training integrity triple. Stamp-Certified IP Ledger provides Ed25519-signed Merkle-replicated distributed attribution. Pledge #2260: defensive patent use only.

Liana Banyan Corporation — Inventor: J. Jones — Provisional Patent Application 23 BP085-BP098 CSIA + Campaign Mechanics Cluster — Sonnet 4.6 SEG — June 2026 7 Claim Groups — 44 Claims — Incremental over Provisional Applications 1-22 v01_AA_STYLE: A&A canonical format — shorter title + TOC + heading hierarchy + claim notation